

ASSIGNMENT 4

Nilesh Kumar
CS23BT006

1. 3 Student-

```
insert INTO student VALUES (27470,"Shivam","Physics",98);  
SELECT * FROM student;
```

				ID	name	dept_name	tot_cred
☐				00128	Zhang	Comp. Sci.	102
☐				12345	Shankar	Comp. Sci.	32
☐				19991	Brandt	History	80
☐				23121	Chavez	Finance	110
☐				27470	Shivam	Physics	98
☐				44553	Peltier	Physics	56
☐				45678	Levy	Physics	46

takes-

```
insert INTO takes VALUES ("00128","CS-190","1","Spring",2017,"A");  
SELECT * FROM takes;
```

					ID	course_id	sec_id	semester	year	grade
☐					00128	CS-101	1	Fall	2017	A
☐					00128	CS-190	1	Spring	2017	A
☐					00128	CS-347	1	Fall	2017	A-
☐					12345	CS-101	1	Fall	2017	C
☐					12345	CS-190	2	Spring	2017	A

Course-

```
insert INTO course VALUES ("CS-270","Intro to Graphics","Comp. Sci.",6);  
SELECT * FROM course;
```

☐	 Edit	 Copy	 Delete	BIO-101	Intro. to Biology	Biology	4
☐	 Edit	 Copy	 Delete	BIO-301	Genetics	Biology	4
☐	 Edit	 Copy	 Delete	BIO-399	Computational Biology	Biology	3
☐	 Edit	 Copy	 Delete	CS-101	Intro. to Computer Science	Comp. Sci.	4
☐	 Edit	 Copy	 Delete	CS-190	Game Design	Comp. Sci.	4
☐	 Edit	 Copy	 Delete	CS-270	Intro to Graphics	Comp. Sci.	6
☐	 Edit	 Copy	 Delete	CS-315	Robotics	Comp. Sci.	3
☐	 Edit	 Copy	 Delete	CS-319	Image Processing	Comp. Sci.	3

Classroom-

```
INSERT INTO classroom VALUES ("Warner",102,500);
SELECT * FROM classroom;
```

+ Options

				building	room_number	capacity
☐	 Edit	 Copy	 Delete	Packard	101	500
☐	 Edit	 Copy	 Delete	Painter	514	10
☐	 Edit	 Copy	 Delete	Taylor	3128	70
☐	 Edit	 Copy	 Delete	Warner	102	500
☐	 Edit	 Copy	 Delete	Watson	100	30
☐	 Edit	 Copy	 Delete	Watson	120	50


 Check all
  With selected
  Edit
  Copy
  Delete
  Export

Department-

```
INSERT INTO department VALUES ("Dance","Warner",85000.00);
SELECT * FROM department;
```

				dept_name	building	budget
☐		Edit		Copy		Delete
				Biology	Watson	90000.00
☐		Edit		Copy		Delete
				Comp. Sci.	Taylor	100000.00
☐		Edit		Copy		Delete
				Dance	Warner	85000.00
☐		Edit		Copy		Delete
				Elec. Eng.	Taylor	85000.00
☐		Edit		Copy		Delete
				Finance	Painter	120000.00
☐		Edit		Copy		Delete
				History	Painter	50000.00
☐		Edit		Copy		Delete
				Music	Packard	80000.00
☐		Edit		Copy		Delete
				Physics	Watson	70000.00

Instructor-

```
INSERT INTO instructor VALUES ("22472","Solanki","Comp. Sci.",66000.00);
SELECT * FROM instructor;
```

				ID	name	dept_name	salary
☐		Edit		Copy		Delete	
				10101	Srinivasan	Comp. Sci.	65000.00
☐		Edit		Copy		Delete	
				12121	Wu	Finance	90000.00
☐		Edit		Copy		Delete	
				15151	Mozart	Music	40000.00
☐		Edit		Copy		Delete	
				22222	Einstein	Physics	95000.00
☐		Edit		Copy		Delete	
				22472	Solanki	Comp. Sci.	66000.00
☐		Edit		Copy		Delete	
				32343	El Said	History	60000.00
☐		Edit		Copy		Delete	
				33456	Gold	Physics	87000.00
☐		Edit		Copy		Delete	
				45565	Katz	Comp. Sci.	75000.00
☐		Edit		Copy		Delete	
				58583	Califieri	History	62000.00
☐		Edit		Copy		Delete	
				76543	Singh	Finance	80000.00
☐		Edit		Copy		Delete	
				76766	Crick	Biology	72000.00

Prereq-

```
INSERT INTO prereq VALUES ("CS-270","CS-101");
SELECT * FROM prereq;
```

+ Options

←

→

▼

course_id

prereq_id

</

Section-

```
INSERT INTO section VALUES ("CS-270",1,"Summer",2017,"Taylor",3128,"D");
SELECT * FROM section;
```

+ Options													
← T →			▼	course_id	sec_id	semester	year	building	room_number	time_slot_id			
☐		Edit		Copy		Delete	BIO-101	1	Summer	2017	Painter	514	B
☐		Edit		Copy		Delete	BIO-301	1	Summer	2018	Painter	514	A
☐		Edit		Copy		Delete	CS-101	1	Fall	2017	Packard	101	H
☐		Edit		Copy		Delete	CS-101	1	Spring	2018	Packard	101	F
☐		Edit		Copy		Delete	CS-190	1	Spring	2017	Taylor	3128	E
☐		Edit		Copy		Delete	CS-190	2	Spring	2017	Taylor	3128	A
☐		Edit		Copy		Delete	CS-270	1	Summer	2017	Taylor	3128	D
☐		Edit		Copy		Delete	CS-315	1	Spring	2018	Watson	120	D
☐		Edit		Copy		Delete	CS-319	1	Spring	2018	Watson	100	B
☐		Edit		Copy		Delete	CS-319	2	Spring	2018	Taylor	3128	C
☐		Edit		Copy		Delete	CS-347	1	Fall	2017	Taylor	3128	A

Teaches-

```
INSERT INTO teaches VALUES ("10101","CS-319",2,"Spring",2018);
SELECT * FROM `teaches`
```

☐		Edit		Copy		Delete	83821	CS-190	1	Spring	2017
☐		Edit		Copy		Delete	83821	CS-190	2	Spring	2017
☐		Edit		Copy		Delete	10101	CS-315	1	Spring	2018
☐		Edit		Copy		Delete	45565	CS-319	1	Spring	2018
☐		Edit		Copy		Delete	10101	CS-319	2	Spring	2018
☐		Edit		Copy		Delete	83821	CS-319	2	Spring	2018

Time_slot-

```
INSERT into time_slot VALUES ("H","W",16,0,16,50);
SELECT * FROM `time_slot`
```

☐		Edit		Copy		Delete	G	M	16	0	16	50
☐		Edit		Copy		Delete	G	W	16	0	16	50
☐		Edit		Copy		Delete	H	W	10	0	12	30
☐		Edit		Copy		Delete	H	W	16	0	16	50

☐ Check all With selected: Edit Copy Delete Export

Advisor-

```
insert INTO advisor VALUES ("27470","10101");
SELECT * FROM `advisor`
```

+ Options						s_ID	i_ID	
↔T↔								
☐		Edit		Copy		Delete	12345	10101
☐		Edit		Copy		Delete	27470	10101
☐		Edit		Copy		Delete	44553	22222
☐		Edit		Copy		Delete	45678	22222
☐		Edit		Copy		Delete	00128	45565

Q2

A.

```
SELECT temp.ID, temp.name, course.course_id, course.title, course.dept_name
from (SELECT t.course_id, i.ID, i.name from instructor as i, teaches as t
where i.ID = t.ID and i.dept_name = "Comp. Sci." and t.year = 2009) as
temp, course where temp.course_id = course.course_id and dept_name = "Civil";
```

☐ Show all | Number of rows: 25 ▼ Filter rows

+ Options

ID	name	course_id	title	dept_name
45565	Katz	CE-101	Intro to Civil	Civil

☐ Show all | Number of rows: 25 ▼ Filter rows

B.

```
INSERT INTO course VALUES ("CS-333","Advanced DBMS","Comp. Sci.",4);
```

☐				CS-270	Intro to Graphics	Comp. Sci.	6
☐				CS-315	Robotics	Comp. Sci.	3
☐				CS-319	Image Processing	Comp. Sci.	3
☐				CS-333	Advanced DBMS	Comp. Sci.	4
☐				CS-347	Database System Concepts	Comp. Sci.	3
☐				EE-181	Intro. to Digital Systems	Elec. Eng.	3

```
INSERT INTO course VALUES ("CS-303","Intro to DBMS","Comp. Sci.",4);
```

```
INSERT into prereq VALUES ("CS-333","CS-303");
```

☐				CS-333	CS-303
☐				EE-181	PHY-101

C.

Before salary

☐				22222	Einstein	Physics	95000.00
☐				22471	Himesh	Civil	68000.00

```
UPDATE instructor set salary = salary*1.1 where ID in (select temp.ID from
(SELECT ID from department as d, instructor as i where d.dept_name =
i.dept_name and budget > 900000) as temp);
```

After increase

☐	 Edit	 Copy	 Delete	22222	Einstein	Physics	95000.00
☐	 Edit	 Copy	 Delete	22471	Himesh	Civil	74800.00
☐	 Edit	 Copy	 Delete	22472	Solanki	Comp. Sci.	66000.00

D.

I am using year 2017 in place of 2007, number of student as 1 instead of 15 so that result is not empty set

```
SELECT temp.course_id, title, cnt from (SELECT t.course_id, count(course_id)
as cnt from student as s, takes as t where s.ID = t.ID and dept_name =
"Comp. Sci." and year = 2017 and semester = "Fall" group by course_id) as
temp , course where temp.course_id = course.course_id and cnt > 1 order by
temp.course_id asc;
```

course_id	1	title	cnt
CS-101		Intro. to Computer Science	4
CS-347		Database System Concepts	2

Q3

Time is not enough for creating rows for year 2010 or 2011, so instead i am using year = 2017 or year = 2018

i.

```
CCREATE view facload as (SELECT name, count(name) as course_no, sum(credits)
as tot_credits FROM course as c, (select i.ID, t.course_id, i.name from
instructor as i , teaches as t where i.ID = t.ID and year in (2017,2018)) as
temp where c.course_id = temp.course_id group by name );
```

```
select * from facload;
```

name	course_no	tot_credits
Crick	2	8
Srinivasan	4	13
Katz	2	7
Brandt	3	11
Kim	1	3
Wu	1	3
El Said	1	3
Mozart	1	3
Einstein	1	4

ii.

```
SELECT f.name, course_no, tot_credits, dept_name from facload as f,
instructor as i where f.name = i.name order by f.tot_credits desc LIMIT 1;
```

name	course_no	tot_credits	1	dept_name
Srinivasan	4	13		Comp. Sci.

Q5.

i.

```
insert into classroom VALUES ("Watson",122,100);
SELECT * FROM `classroom`
```


← T →				building	room_number	capacity
☐	 Edit	 Copy	 Delete	Packard	101	500
☐	 Edit	 Copy	 Delete	Painter	514	10
☐	 Edit	 Copy	 Delete	Taylor	3128	70
☐	 Edit	 Copy	 Delete	Warner	102	500
☐	 Edit	 Copy	 Delete	Watson	100	30
☐	 Edit	 Copy	 Delete	Watson	120	50
☐	 Edit	 Copy	 Delete	Watson	121	40
☐	 Edit	 Copy	 Delete	Watson	122	100